Optimization-based Invariant Generation for Cyber-Physical Systems

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PhD Defense
May 7, 2026

Contents

What is an invariant?

In mathematics, an invariant is a property of a mathematical object which remains unchanged after operations or transformations.

In computer science, an invariant is a logical assertion that is always held to be true during a certain phase of execution of a computer program.

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Verifying Program Correctness

- Consider a piece of program:

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&\text{while } x \leq 100 \text{ do}\\
&\quad x \leftarrow 2x - 1\\
&\text{end}
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- Question: How to reason about the correctness of a program?

To prove that $x > 0$ always holds.

- Raised by von Neumann (1946), Turing (1949), and McCarthy (1960).
- Formalized by Floyd, Hoare, and Naur in 1960s. $\Longrightarrow$ **Hoare logic**.

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Invariants and Inductive Invariants

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- **Loop Invariant**: An assertion at the loop entrance that holds whenever the program reaches that point (e.g. $x > 0$).
- **Loop Inductive Invariant**: A loop invariant that is inductive, i.e., if it holds in one iteration, then it still holds in the next iteration. (e.g. $x \geq 1$).

$$x \geq 1 \land x' = 2x - 1 \land x \leq 100 \implies x' \geq 1$$

- By "invariants", we always mean inductive invariants.

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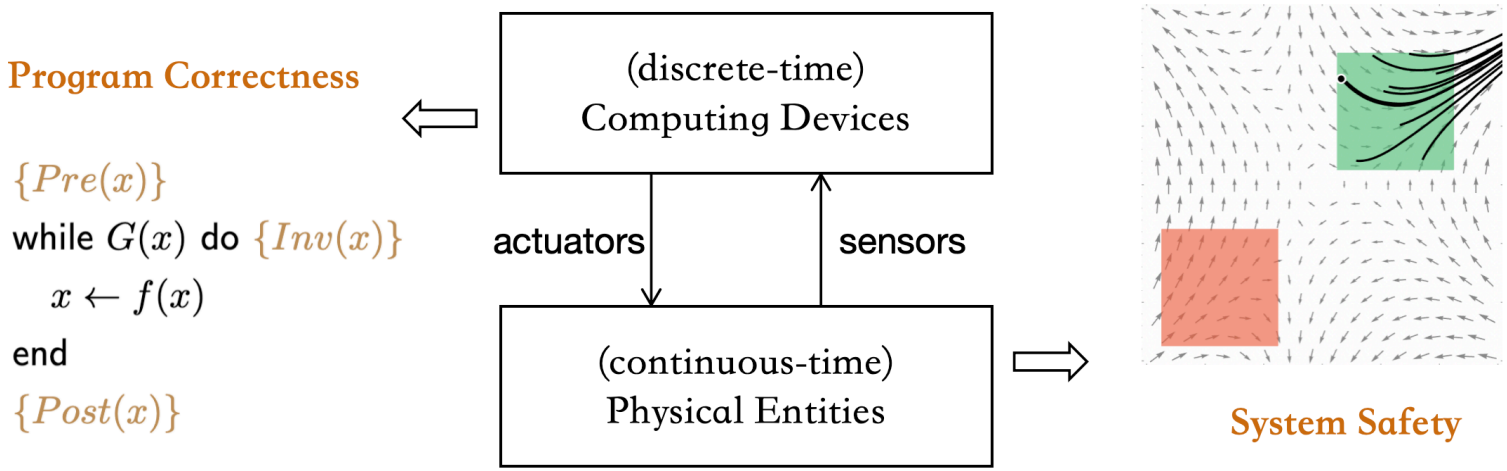
Verification Conditions

$\{Pre(x)\}$
while $G(x)$ do $\{Inv(x)\}$
  $x \leftarrow f(x)$
end
$\{Post(x)\}$

- Proving loop correctness is reduced to finding an suitable invariant:

$$\forall x.\ Pre(x) \implies Inv(x)$$
$$\forall x.\ Inv(x) \wedge G(x) \implies Inv(f(x))$$
$$\forall x.\ Inv(x) \wedge \neg G(x) \implies Post(x)$$

Safety of Cyber-Physical Systems (CPS)



- **CPS** = embedded system + deep integration of computing, communication, and control
- **Safety**: the system will not evolve into bad states.
- Focus of this thesis: a unifed framework for both discrete and continuous dynamics.

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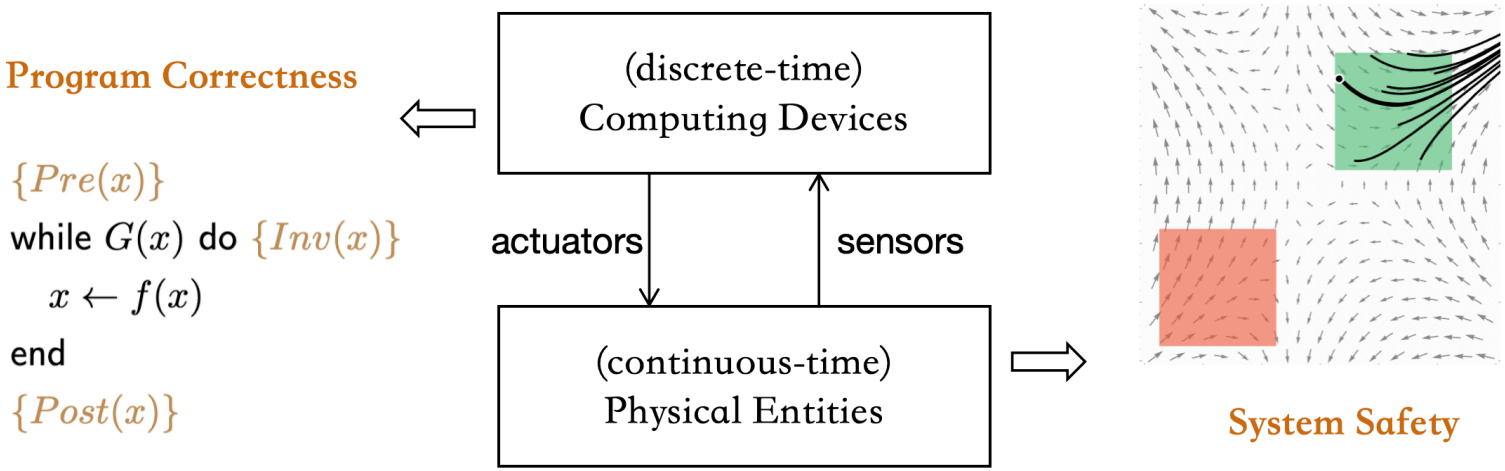

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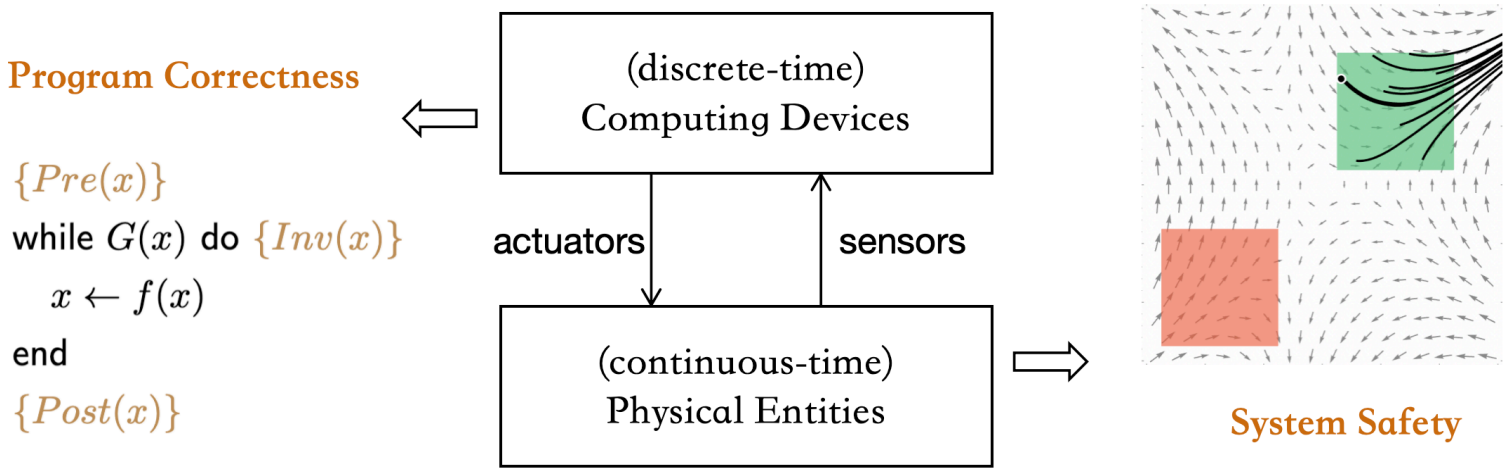

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Extending to Continuous-Time Dynamical Systems

System dynamics given by Ordinary Differential Equations (ODEs):

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}), \quad \boldsymbol{x} \in \mathcal{X}_d$$

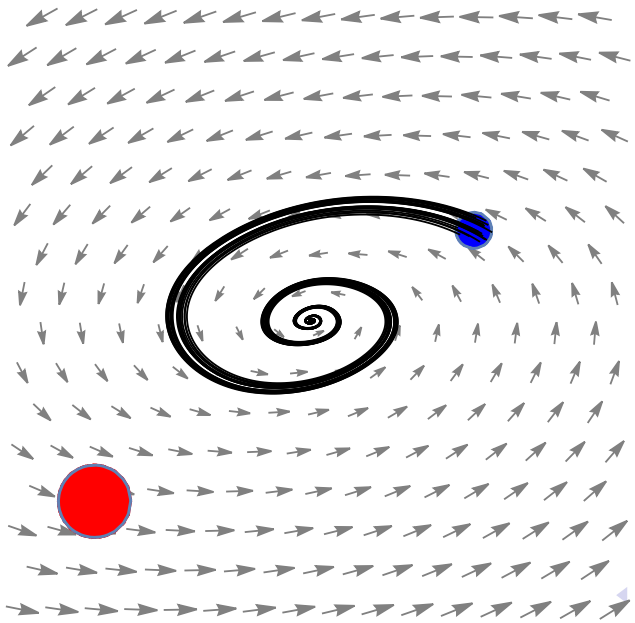
- $\boldsymbol{f}: \mathbb{R}^n \to \mathbb{R}^n$ is a locally Lipschitz continuous vector field.
- Unique trajectory $\xi_{\boldsymbol{x}_0} : \mathbb{R}_{\geq 0} \to \mathbb{R}^n$ for any initial state.

Safety Verification Problem

Safety Verification Problem of ODE Systems: Given domain $\mathcal{X}_d \subseteq \mathbb{R}^n$, initial set $\mathcal{X}_i \subset \mathcal{X}_d$, and unsafe set $\mathcal{X}_u \subset \mathcal{X}_d$ such that $\mathcal{X}_u \cap \mathcal{X}_i = \emptyset$. Prove that

$$\mathcal{X}_u \text{ is not reachable from any state in } \mathcal{X}_i.$$

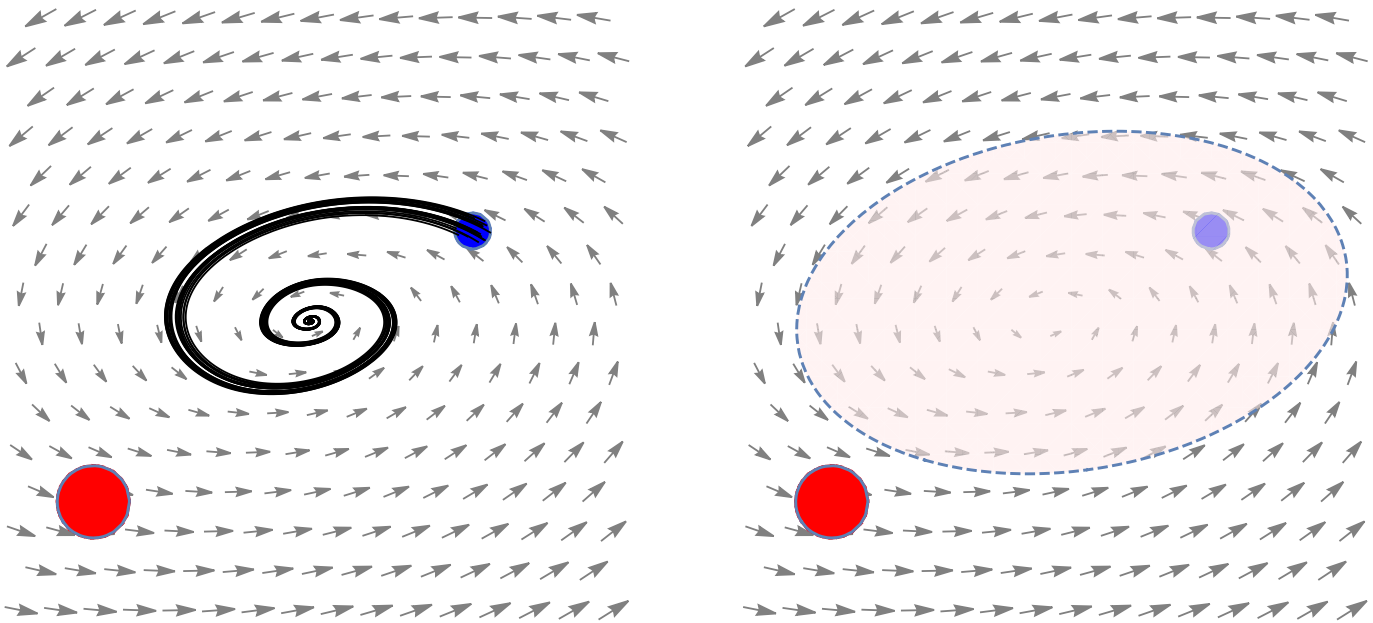
- $\mathcal{X}_i$: pre-condition.
- $\mathcal{X}_u$: negation of post-condition.
- $\mathcal{X}_d$: variable range



Positive Invariants

- Analogous to **inductive invariants** for programs.
- A **positive/differential invariant** is a subset of states $\Omega \subseteq \mathcal{X}_d$ s.t. any trajectory starting from Ω stays within Ω forever, i.e.,

$$\forall \mathbf{x}_0 \in \Omega, \forall t \in \mathbb{R}_{\geq 0}.\ \xi_{\mathbf{x}_0}(t) \in \Omega.$$



Research Question

- The invariant generation problem is undecidable in general.
  - Because finding the exact set of reachable states is undecidable.
- Undecidable even for affine programs with affine guards [Müller-Olm et al., 2004].
- Undecidable even for polynomial ODE [Graça et al., 2008].

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Invariant Generation Methods

- **Programs**:
  - Constraint Solving
  - Abstract Interpretation
  - Recursive Relation
  - Counter-Example Guided Inductive Synthesis
  - Machine Learning
- **Continuous-time systems**:
  - Constraint Solving
  - Differential Algebra
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- **Advantages** of Constraint Solving:
  - Works for both discrete- and continuous-time systems.
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Assumptions

(1) Variables are $\mathbb{R}$-valued.
- No integer/Boolean variables. No other data structures.

(2) Functions are polynomial:
- program updates and ODE dynamics.

(3) Predicates are basic semialgebraic sets
- target invariants, pre- and post-conditions, initial, unsafe, and domain regions
- defined by a conjunction of polynomial inequalities

$$S = \{\boldsymbol{x} \in \mathbb{R}^n \mid p_1(\boldsymbol{x}) \geq 0, \ldots, p_m(\boldsymbol{x}) \geq 0\} \quad \text{with } p_i \in \mathbb{R}[\boldsymbol{x}].$$

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Constraint-based synthesis: inductive invariants

1. **Template Setting**: set a parameterized template to represent the invariant $I(\boldsymbol{x}) \leq 0$ where

$$I(\mathbf{x}; \mathbf{a}) = a_1 x_1^2 + a_2 x_1 x_2 + a_3 x_2^2 + a_4 x_1 + a_5 x_2 + a_6 \in \mathbb{R}[x_1, x_2].$$

2. **Condition Encoding**: encode the invariant conditions into constraints.

$$\text{Substitute } Inv(\boldsymbol{x}) \text{ by } I(\boldsymbol{x}; \boldsymbol{a}) \leq 0$$

3. **Constraint Solving**: solve the constraints for the parameters $\boldsymbol{a}$:
   1. $\forall \mathbf{x}.\ \text{Pre}(\boldsymbol{x}) \implies I(\mathbf{x}; \mathbf{a}) \leq 0,$
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   sufficient and necessary condition for $I(\boldsymbol{x}, \boldsymbol{a}) \leq 0$ to be an inductive invariant.

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Constraint-based synthesis: positive invariants

- Set a template $B(\boldsymbol{x};\boldsymbol{a}) \leq 0$ for the positive invariant.
  - B is called a **Barrier Certificate (BC)** [Prajna et al., 2004].
- Encode the sufficient conditions for positive invariance:
  1. $\forall \mathbf{x}.\ \mathbf{x} \in \mathcal{X}_i \implies B(\mathbf{x};\mathbf{a}) \leq 0,$
  2. $\forall \mathbf{x}.\ \mathbf{x} \in \mathcal{X}_d \wedge B(\mathbf{x};\mathbf{a}) = 0 \implies \mathfrak{L}_{\boldsymbol{f}} B(\mathbf{x};\mathbf{a}) < 0,$
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From the view of constraint solving

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Tools: SOS polynomials

- A polynomial $\sigma(\boldsymbol{x})$ is **Sum-of-Squares** (SOS) if it can written as

$$\sigma(\boldsymbol{x}) = \sum_{i=1}^{n} p_i^2(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}].$$

- $\Sigma[\boldsymbol{x}]$ denotes the set of SOS polynomials.
- A polynomial $\sigma(\boldsymbol{x})$ is SOS iff the Gram matrix of σ is PSD.

$$\begin{aligned}
\sigma(x_1, x_2) &= 2x_1^4 + 2x_1^3x_2 - x_1^2x_2^2 + 5x_2^4 \\
&= \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1x_2 \end{pmatrix}'}_{\text{basis}} \underbrace{\begin{pmatrix} 2 & -3 & 1 \\ -3 & 5 & 0 \\ 1 & 0 & 5 \end{pmatrix}}_{\text{Gram matrix}} \underbrace{\begin{pmatrix} x_1^2 \\ x_2^2 \\ x_1x_2 \end{pmatrix}}_{\text{basis}} \\
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Tools: Putinar's Positivstellensatz

- A basic semialgebraic set $S = \{x \mid p_i(x) \geq 0,\ i = 1, \ldots, m\}$.
- A **Quadratic Module** associated to S:

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  - This basically means that S is bounded.

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SOS Constraints for Inductive Invariants

- For simplicity, we assume that $\mathrm{Pre}(\boldsymbol{x})$ is $p(\boldsymbol{x}) \leq 0$ and $\mathrm{Post}(\boldsymbol{x})$ is $q(\boldsymbol{x}) \leq 0$:
  1. $\forall \mathbf{x}.\ p(\mathbf{x}) \leq 0 \implies I(\mathbf{x}; \mathbf{a}) \leq 0,$
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- By applying **Putinar's Psatz**, the above constraints are satisfiable if the following constraints are solvable, for any $\epsilon > 0$ arbitrarily small.

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&\text{find} \quad \boldsymbol{a}, \boldsymbol{b} \\
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Three challenges in optimization-based invariant generation:

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Weak Invariants vs Strong Invariants

- Given an invariant template $I(\boldsymbol{x}; \boldsymbol{a})$.
- The **weak** invariant generation problem asks for one invariant satisfying the template, i.e., to find a valid parameter assignment.
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- The quantified constraints:

$$\forall \mathbf{x}.\ \bigwedge_{i=1}^{m} k_i(\mathbf{x};\mathbf{a}) \geq 0 \implies l(\mathbf{x};\mathbf{a}) \geq 0$$

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$$R = \{\mathbf{a} \mid \phi(\mathbf{a}) \leq 0\}.$$

where ϕ can be proved to be upper-semicontinuous, i.e., can be approximated from above by polynomials.

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  - Allow the template $I(\boldsymbol{x}; \boldsymbol{a})$ to be nonlinear in $\boldsymbol{a}$, or even a parameterized formula.
  - Soundness, convergence, and weak completeness guarantee.
- **Cons**
  - SDP constraint size is polynomial in $|\boldsymbol{a}| + |\boldsymbol{x}|$.
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- When R has zero-measure, for a special subclass of templates, we give a sound and complete SDP-based algorithm by using variable substitution:
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- For systems with unbounded domains, encoding SOS constraints is sound but not relatively complete
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Algebraic Tool: Homogenization

- Introduce a fresh variable x_0. Denote $\tilde{\mathbf{x}} = (x_0, \mathbf{x}) \in \mathbb{R}^{n+1}$.
- Let $p(\mathbf{x})$ be a polynomial of degree d, its **homogenization** is

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- "$S = \tilde{S}_+$": an one-to-one projection

$$\mathbf{x} \in S \iff \left(\frac{1}{\sqrt{1 + \|\mathbf{x}\|^2}}, \frac{x_1}{\sqrt{1 + \|\mathbf{x}\|^2}}, \ldots, \frac{x_n}{\sqrt{1 + \|\mathbf{x}\|^2}} \right) \in \tilde{S}_+.$$

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- "$S = \tilde{S}_+$": an one-to-one projection

$$\mathbf{x} \in S \iff \left(\frac{1}{\sqrt{1+\|\mathbf{x}\|^2}}, \frac{x_1}{\sqrt{1+\|\mathbf{x}\|^2}}, \ldots, \frac{x_n}{\sqrt{1+\|\mathbf{x}\|^2}}\right) \in \tilde{S}_+.$$

- closure$(\tilde{S}_+) = \tilde{S}$

Relation between S and $\tilde{S}$

$$
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S &= \{\mathbf{x} \in \mathbb{R}^n \mid p_1(\mathbf{x}) \geq 0, \ldots, p_m(\mathbf{x}) \geq 0\},\\
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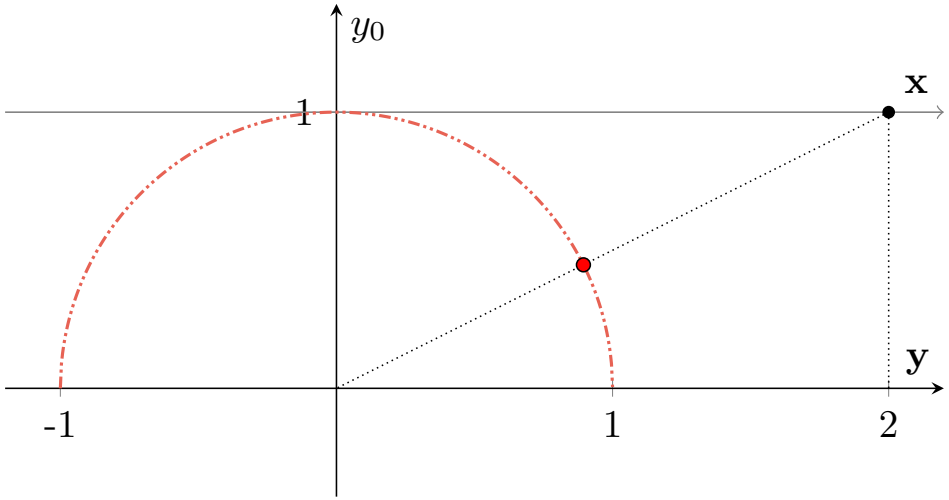
Geometric Explanation for $\mathbb{R}^1$



$$x \in \mathbb{R} \longleftrightarrow (y_0, y) \hat{=} (\underbrace{\frac{1}{\sqrt{1+x^2}}}_{x_0}, \underbrace{\frac{x}{\sqrt{1+x^2}}}_{\text{new } x}) \in \tilde{S}_+$$

- Points with $x_0 = 0$ in $\tilde{S}$ correspond to **points at infinity** in $\mathbb{R}^n$.
- We need a technical assumption for these points.

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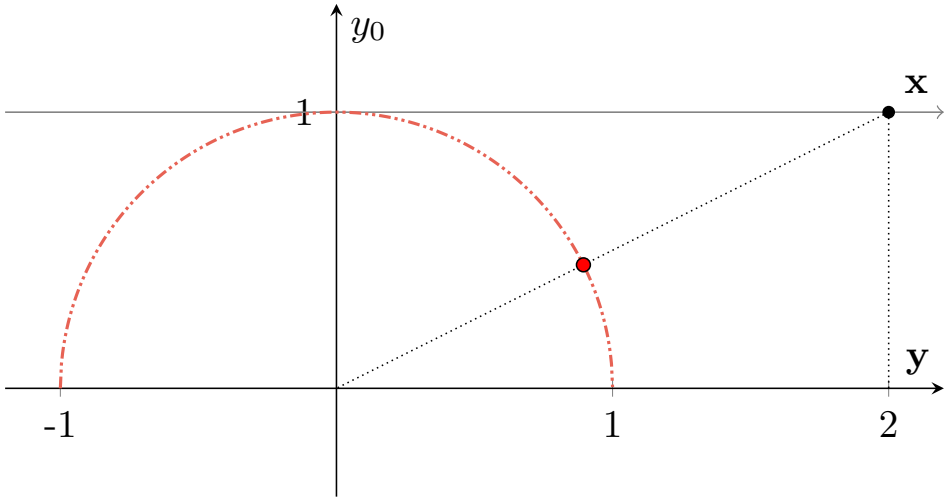

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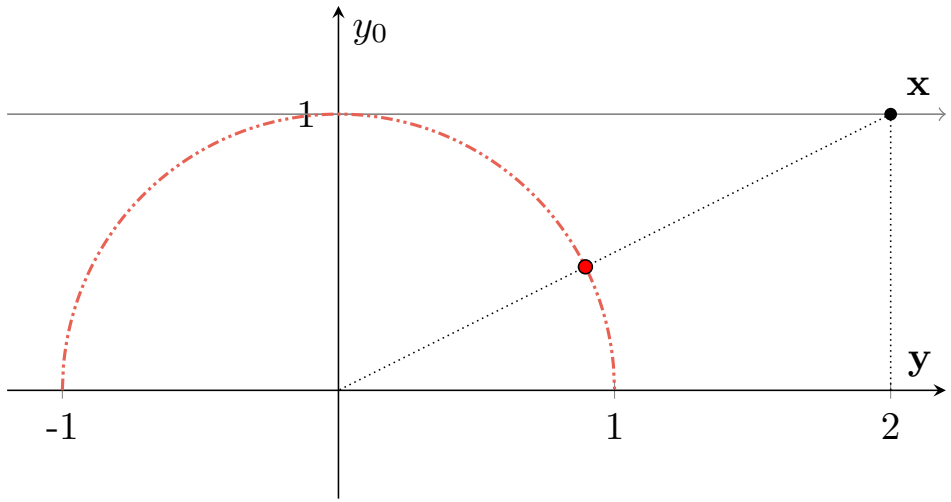

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Closed at infinity

Definition (closed at infinity [Nie, 2012])

A basic semialgebraic set S is closed at ∞ if

$$\text{closure}(\tilde{S}_+) = \tilde{S}.$$

- Depends on the description

$$S_1 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0\} \text{ not closed at } \infty$$

$$S_2 = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 - x_2^2 \geq 0, \underbrace{x_1 \geq 0}_{\text{redundant}}\} \text{ closed at } \infty$$

Homogenization: Main Theorem

Theorem (Equivalence [Huang et al., 2023])

When S is closed at ∞,

$$f(\mathbf{x}) \geq 0 \text{ over } S \iff \tilde{f}(\tilde{\mathbf{x}}) \geq 0 \text{ over } \tilde{S}$$

- $\tilde{S}$ lies on the half unit sphere, hence bounded.
- The quadratic module $\mathbf{QM}(\tilde{S})$ becomes Archimedean.

Applied for (differential) invariants

- Original SOS constraints (only sound), for any $\epsilon > 0$:

$$
\begin{aligned}
&-B(\mathbf{x}) \in \mathbf{QM}(\mathcal{X}_i),\\
&B(\mathbf{x}) - \epsilon \in \mathbf{QM}(\mathcal{X}_u)\\
&\lambda(\mathbf{x})B(\mathbf{x}) - \mathfrak{L}_{\boldsymbol{f}}B(\mathbf{x}) \in \mathbf{QM}(\mathcal{X}_d)
\end{aligned}
$$

- First **homogenization**, then **Putinar's Psatz**.

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\begin{aligned}
&-\tilde{B}(x_0,\mathbf{x}) \in \mathbf{QM}(\tilde{\mathcal{X}}_i)\\
&\tilde{B}(x_0,\mathbf{x}) - \epsilon x_0^d \in \mathbf{QM}(\tilde{\mathcal{X}}_u)\\
&\tilde{H}(x_0,\mathbf{x}) \in \mathbf{QM}(\tilde{\mathcal{X}}_d)
\end{aligned}
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  where $H(\mathbf{x}) = \lambda(\mathbf{x})B(\mathbf{x}) - \mathfrak{L}_{\boldsymbol{f}}B(\mathbf{x})$.

- Assume that $\mathcal{X}_i$, $\mathcal{X}_u$, and $\mathcal{X}_d$ are all closed at infinity. Under some mild conditions (ϵ can take 0), both sound and relatively complete.

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Semialgebraic Barrier Certificates

- Set a template for $B(\boldsymbol{x}) \in \mathbb{R}[\boldsymbol{x}]$, then compute $\tilde{B}(x_0, \boldsymbol{x})$.
- Alternatively, we can set a polynomial template for $\tilde{B}(x_0, \boldsymbol{x})$, then the original $B(\frac{1}{\sqrt{\|\boldsymbol{x}\|^2+1}}, \frac{\boldsymbol{x}}{\sqrt{\|\boldsymbol{x}\|^2+1}})$ can be non-polynomial:

$$(\sqrt{\|\boldsymbol{x}\|^2+1})^d \cdot B(\frac{1}{\sqrt{\|\boldsymbol{x}\|^2+1}}, \frac{\boldsymbol{x}}{\sqrt{\|\boldsymbol{x}\|^2+1}})$$
$$= \underbrace{B_1(\boldsymbol{x}) + \sqrt{\|\boldsymbol{x}\|^2+1} \cdot B_2(\boldsymbol{x})}_{\text{semialgebraic BC}} \quad B_1, B_2 \in \mathbb{R}[\boldsymbol{x}]$$

- Semialgebraic BC subsumes polynomial BC as a special case, where $B_2(\boldsymbol{x})$ is constant zero.

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Experiments

			Incomplete			Polynomial BC			Semialgebraic BC		
system	dim	unbounded	deg	succ	time(s)	deg	succ	time(s)	deg	succ	time(s)
vector[37]	2	$\mathcal{X}$	4	✓	0.58	3	✓	0.03	**2**	✓	0.83
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.39	4	✓	0.16	**2**	✓	0.72
barrier[29]	2	$\mathcal{X}$	>6	✗	2.40	>6	✗	2.73	>4	✗	22.98
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.78	3	✓	0.04	**2**	✓	3.12
lie-der[24]	2	$\mathcal{X}$	3	✓	0.19	3	✓	0.14	**1**	✓	0.75
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**3**	✓	0.14	**3**	✓	0.19	**3**	✓	5.04
arch1[36]	2	$\mathcal{X}$	**4**	✓	0.40	**4**	✓	0.54	>4	✗	117.61
		$\mathcal{I},\mathcal{U},\mathcal{X}$	**1**	✓	0.09	**1**	✓	0.04	2	✓	20.92
arch2[36]	2	$\mathcal{X}$	**3**	✓	0.20	**3**	✓	0.21	**3**	✓	5.53
		$\mathcal{I},\mathcal{U},\mathcal{X}$	3	✓	0.18	3	✓	0.18	**2**	✓	1.59
arch3[36]	2	$\mathcal{X}$	**2**	✓	0.12	**2**	✓	0.13	**2**	✓	3.45
		$\mathcal{I},\mathcal{U},\mathcal{X}$	>6	✗	0.84	>6	✗	1.30	**1**	✓	0.34
arch4[36]	2	$\mathcal{X}$	>6	✗	0.11	5	✓	0.74	**3**	✓	5.69
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**6**	✓	1.21	>4	✗	7.74
nagumo[34]	2	$\mathcal{X}$	**2**	✓	0.13	**2**	✓	0.14	**2**	✓	3.50
		$\mathcal{U},\mathcal{X}$	>6	✗	1.02	**3**	✓	0.17	>4	✗	23.73
lorenz[10]	3	$\mathcal{X}$	6	?	TO	**4**	✓	72.13	2	?	TO
		$\mathcal{U},\mathcal{X}$	**5**	✓	248.88	6	?	TO	2	?	TO
lotka[14]	3	$\mathcal{X}$	>6	✗	88.97	>6	✗	27.8	3	?	TO
		$\mathcal{U},\mathcal{X}$	>6	✗	0.27	>6	✗	438.54	3	?	TO

- Expressiveness: Semialgebraic $>$ Polynomial $>$ Incomplete
- Efficiency: Incomplete $\approx$ Polynomial $\gg$ Semialgebraic

Solving the Positive Invariant Constraints

$$\forall \boldsymbol{x} \in \mathcal{X}_d.\ B(\boldsymbol{x}) = 0 \implies \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) < 0.$$

$$\lambda(\boldsymbol{x}; \boldsymbol{a}) B(\boldsymbol{x}; \boldsymbol{b}) - \mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}; \boldsymbol{b}) \in \mathbf{QM}(\mathcal{X}_d)$$

- When $\lambda(\boldsymbol{x})$ is known, the constraints can be reduced to **LMI**
  - fix $\lambda(\boldsymbol{x})$, solve as SDP. [Kong et al., 2013; Dai et al., 2017]
- In general, **BMI** constraints:
  - [Yang et al., 2015; Chen et al., 2020] alternating descent: efficient, no convergence guarantee
  - [Wang et al., 2021; Wang et al., 2022] **difference-of-convex (DC)**: guarantees to terminate and converge but requires good initial points

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A Closer Look at $\lambda(\boldsymbol{x}; \boldsymbol{a})$

- Idea: choose a "good" $\lambda(\boldsymbol{x}; \boldsymbol{a})$ to attain expressiveness and to simplify the DC procedure.
- First, exponential-type conditions ($\lambda(\boldsymbol{x}; \boldsymbol{a})$ degenerates to a constant) are usually too conservative.

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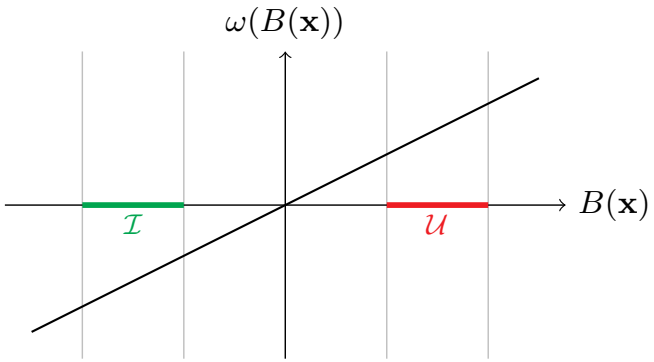
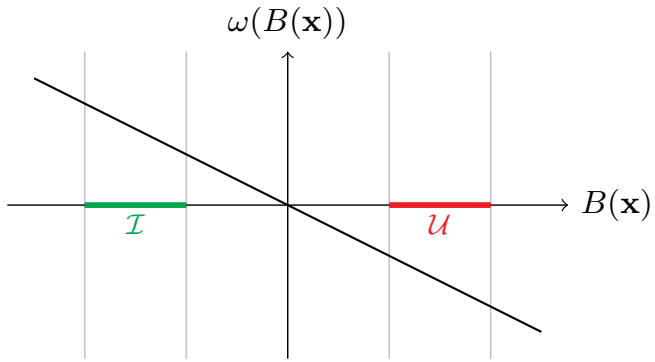


(a) $\omega(b) = \lambda b,\ \lambda > 0$

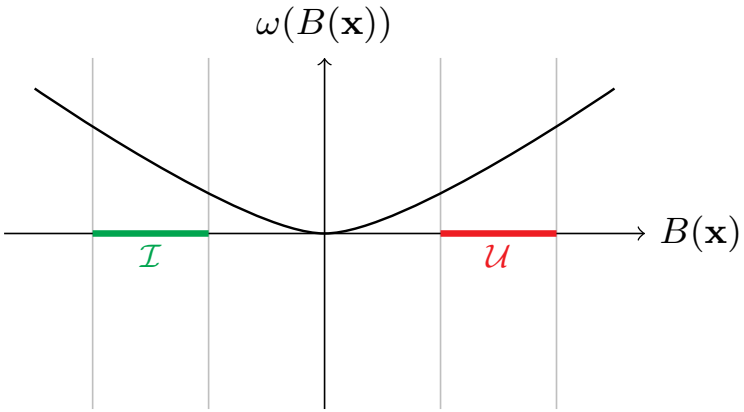


(b) $\omega(b) = \lambda b,\ \lambda < 0$

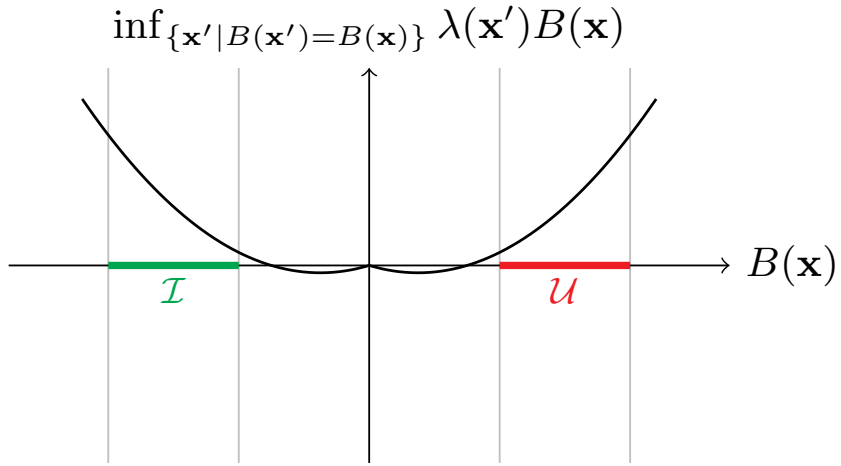
- Left: $\lambda > 0$, $\mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) < 0$ whenever $B(\boldsymbol{x}) < 0$
  $\Longrightarrow$ trajectories must stabilize to the minimizer of $B(\boldsymbol{x})$.
- Right: $\lambda < 0$, $\mathfrak{L}_{\boldsymbol{f}} B(\boldsymbol{x}) < 0$ whenever $B(\boldsymbol{x}) > 0$
  $\Longrightarrow$ trajectories must stabilize to the region $B(\boldsymbol{x}) \leq 0$.

A Closer Look at $\lambda(\boldsymbol{x};\boldsymbol{a})$

- Ideally, we would like to choose $\lambda(\boldsymbol{x};\boldsymbol{a})$ to make $\lambda(\boldsymbol{x})B(\boldsymbol{x})$, denoted $\omega(B(\mathbf{x}))$, positive whenever $B(\mathbf{x}) \neq 0$.



(a) the ideal case for ω



(b) $\lambda(\mathbf{x}) < 0$ over $\mathcal{X}_i$ and > 0 over $\mathcal{X}_u$

- Observation: we can approximate the ideal case by choosing a $\lambda(\boldsymbol{x};\boldsymbol{a}) < 0$ over $\mathcal{X}_i$ and > 0 over $\mathcal{X}_u$.
  - Such $\lambda(\boldsymbol{x})$ is called an **interpolant** between $\mathcal{X}_i$ and $\mathcal{X}_u$.
  - Find some "good" interpolants to serve as DC initial points.

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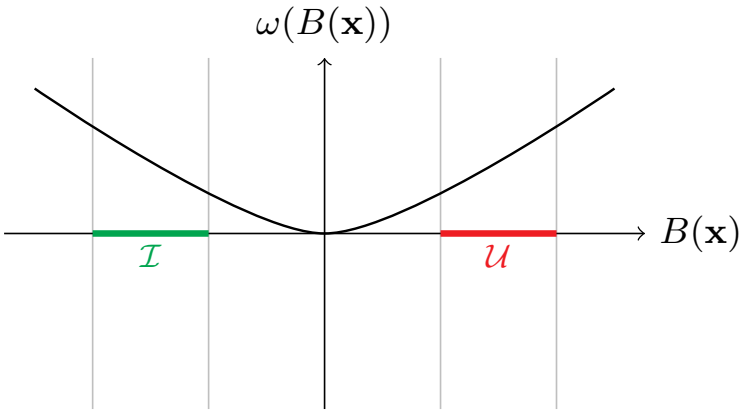

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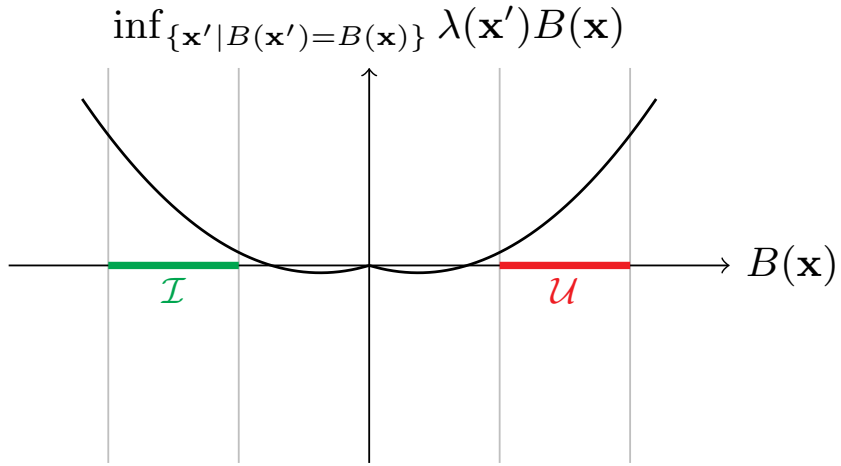

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A Simple Case Study

- The algorithm has been implemented in our toolchain **MARS 2.0**.
- A simple example

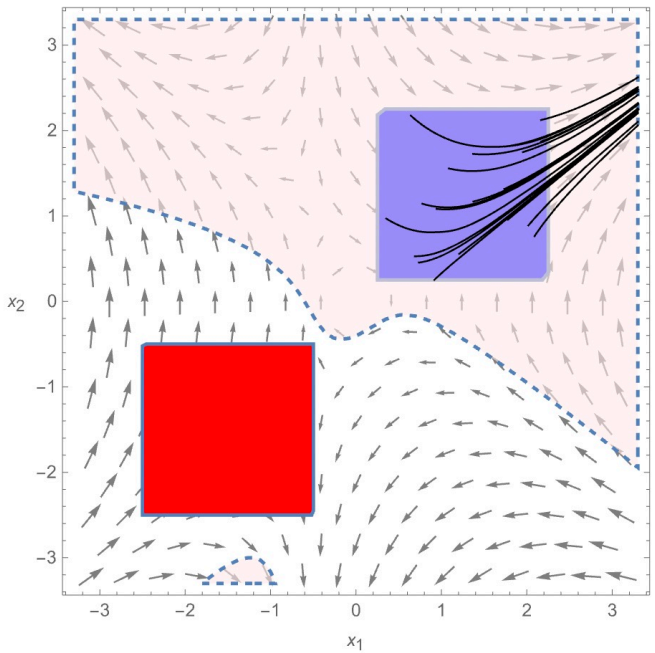
$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \begin{pmatrix} x_2 + 2x_1x_2 \\ -x_1 - x_2^2 + 2x_1^2 \end{pmatrix}$$

with $\mathcal{X}_d = [-3, 3] \times [-3, 3]$, $\mathcal{X}_i = [0.25, 2.25] \times [0.25, 2.25]$, and $\mathcal{X}_u = [-2.5, -0.5] \times [-2.5, -0.5]$.

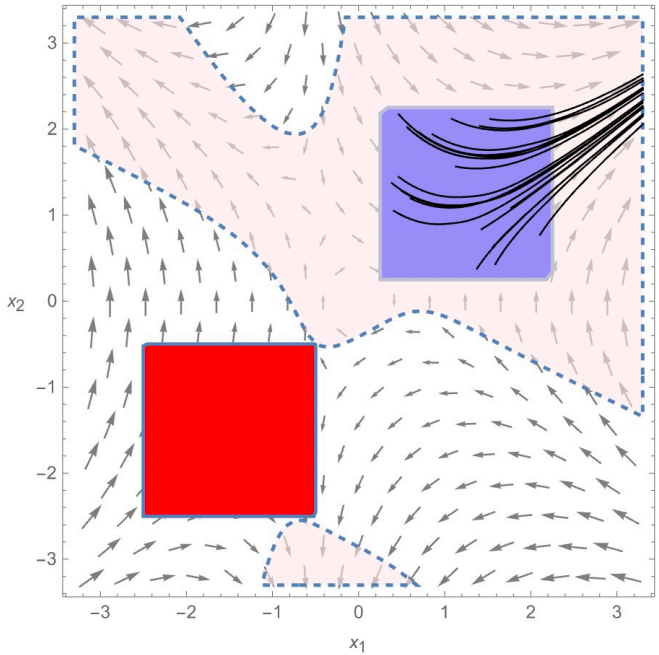
- Using SDP to solve the exponential-type BC condition [Kong et al., 2013] or using alternating descent method to solve non-convex BC condition [Yang et al., 2015] will fail to find a BC of degree $1 \sim 6$.

A Simple Case Study

- Interpolation&DC: find a BC at degree 4, taking 26.4s.
- Original DC: find a BC at degree 5, taking 291.5s.



(a) ItpDC: degree 4



(b) DC: degree 5

Summary

We show how to exploit advanced optimization techniques in the invariant generation problem for both discrete- and continuous-time dynamics:

- **Robust Optimization** techniques for under-approximating strong invariants.
- **Homogenization** techniques for encoding conditions over unbounded domains.
- **Non-Convex Optimization** techniques for solving constraint.

Future Direction: use sparsity to conquer scalability

- Reasoning about nonlinear arithmetic + quantifier alternation is inherently hard.
  - It is unrealistic to expect to find an general and efficient solution.
- In practice, we have these observations:
  - Nonlinearity usually appears locally among some variables.
  - Real invariants (especially for programs) usually contain very few monomials.
- Question: How to exploit such sparsity information in template setting and constraint solving?
- Our recent attempt: combine **treewidth** techniques (sparsity from graph theory) with **quantifier elimination**. $\Longrightarrow$ TACAS'26.

Future Direction: use sparsity to conquer scalability

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Future Direction: a unified framework

In mathematics, an invariant is a property of a mathematical object which remains unchanged after operations or transformations.

- The constraint-based idea provides a way to encoding different types of operations and properties.
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  - **ODE/Hybrid systems**: barrier certificates, Lyapunov functions
  - **Stochastic systems**: stochastic invariants, ranking supermartingales
  - **Control systems**: control barrier functions, control Lyapunov functions
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Experience and Published Papers

UCAS from 2016 to 2026:

- **Hao Wu***, Jiyu Zhu*, Amir Goharshady, Jie An, Bican Xia, and Naijun Zhan. Quantifier Elimination Meets Treewidth. **TACAS'2026**.
- Yihao Yin, **Hao Wu**, Wan Liu, Shuling Wang, Xiong Xu, Wang Lin, Fanjiang Xu, and Naijun Zhan. Formal Design of Safety-Critical Systems with MARs. **JSA'2026**.
- **Hao Wu**, Qiuye Wang, Bai Xue, Naijun Zhan, Lihong Zhi, and Zhi-Hong Yang. Synthesizing Invariants for Polynomial Programs by Semidefinite Programming. **TOPLAS'2025**.
- Xiong Xu, Jean-Pierre Talpin, Shuling Wang, **Hao Wu**, Bohua Zhan, Xinxin Liu, and Naijun Zhan. HpC: A Calculus for Hybrid and Mobile Systems. **OOPSLA'2025**.
- Guanhua Lin, **Hao Wu**, and Naijun Zhan. Barrier Certificate Synthesis via Interpolation and Difference-of-Convex Programming. **Festschrift of Prof. Martin Fränzle's 60th birthday, 2025.**
- **Hao Wu**, Shenghua Feng, Ting Gan, Jie Wang, Bican Xia, and Naijun Zhan. On Completeness of SDP-Based Barrier Certificate Synthesis over Unbounded Domains. **FM'2024**.
- **Hao Wu***, Jie Wang*, Bican Xia, Xiakun Li, Naijun Zhan, and Ting Gan. Nonlinear Craig Interpolant Generation Over Unbounded Domains by Separating Semialgebraic Sets. **FM'2024**.
- Yihao Yin, **Hao Wu**, Shuling Wang, Xiong Xu, Fanjiang Xu, and Naijun Zhan. The Design of Intelligent Temperature Control System of Smart House with MARS. **SETTA'2024**.
- **Hao Wu**, Yu-Fang Chen, Zhilin Wu, Bican Xia, and Naijun Zhan. A decision procedure for string constraints with string/integer conversion and flat regular constraints. **Acta Informatica'2024**.
- 王淑灵, 詹博华, 盛欢欢, **吴昊**, 易士程, 王令泰, 金翔宇, 薛白, 李静辉, 向霜晴, 向展, 毛碧飞. 可信系统性质的分类和形式化研究综述. **软件学报'2022**.
- Xiaohui Bei, Xiaoming Sun, **Hao Wu**†, Jialin Zhang, Zhijie Zhang, and Wei Zi. Cake Cutting on Graphs: A Discrete and Bounded Proportional Protocol. **SODA'2020** (undergraduate period).

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